ADVANCE 5G AND 6G COMMUNICATION SYSTEM

DATE :05.11.26

EXPERIMENT 2

OBJECTIVE 1:

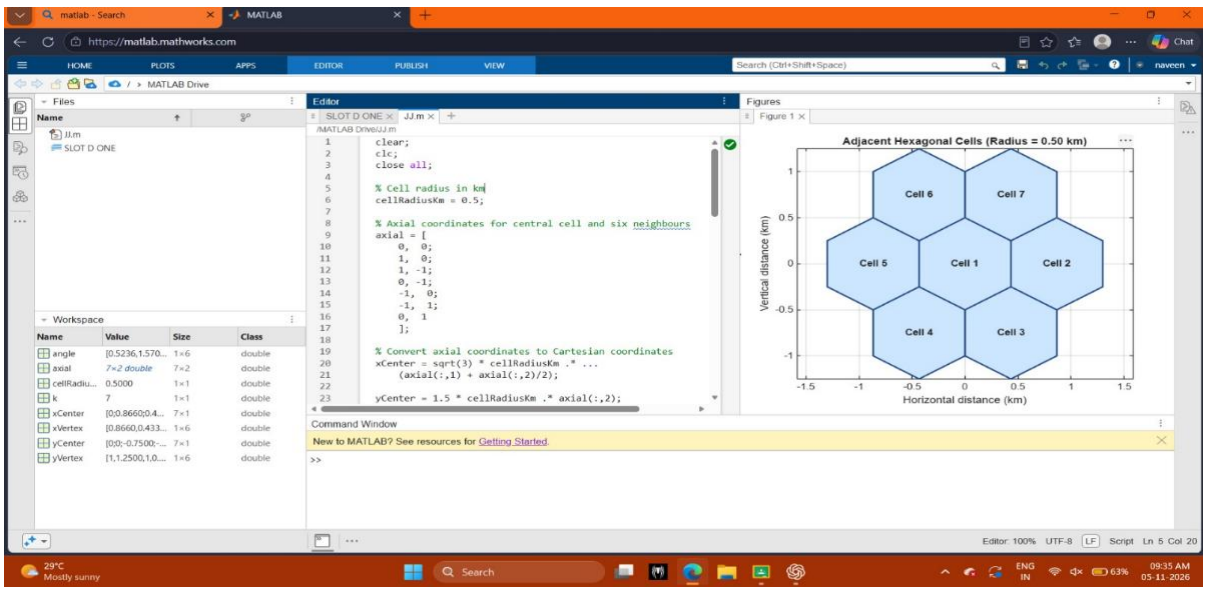
PROGRAM CODE :

```
clear;
clc;
close all;
cellRadiusKm = 0.5;
% Axial coordinates for the central cell and its six neighbours
axial = [ ...
0, 0;
1, 0;
1, -1;
0, -1;
-1, 0;
-1, 1;
0, 1];
% Convert axial coordinates to Cartesian cell-centre positions
xCenter = sqrt(3) * cellRadiusKm .* ...
(axial(:,1) + axial(:,2)/2);
yCenter = 1.5 * cellRadiusKm .* axial(:,2);
% Hexagon vertices
angle = deg2rad(30:60:330);
figure;
hold on;
for k = 1:length(xCenter)
xVertex = xCenter(k) + cellRadiusKm * cos(angle);
```

```
yVertex = yCenter(k) + cellRadiusKm * sin(angle);
patch(xVertex, yVertex, [0.80 0.90 1.00], ...
'EdgeColor', [0.10 0.25 0.50], ...

'LineWidth', 1.5);
text(xCenter(k), yCenter(k), sprintf('Cell %d', k), ...
'HorizontalAlignment', 'center', ...
'FontWeight', 'bold');
end
axis equal;
grid on;
xlabel('Horizontal distance (km)');
ylabel('Vertical distance (km)');
title(sprintf('Adjacent Hexagonal Cells (Radius = %.2f km)', ...
cellRadiusKm));
hold off;
```

EXPECTED OUTPUT :



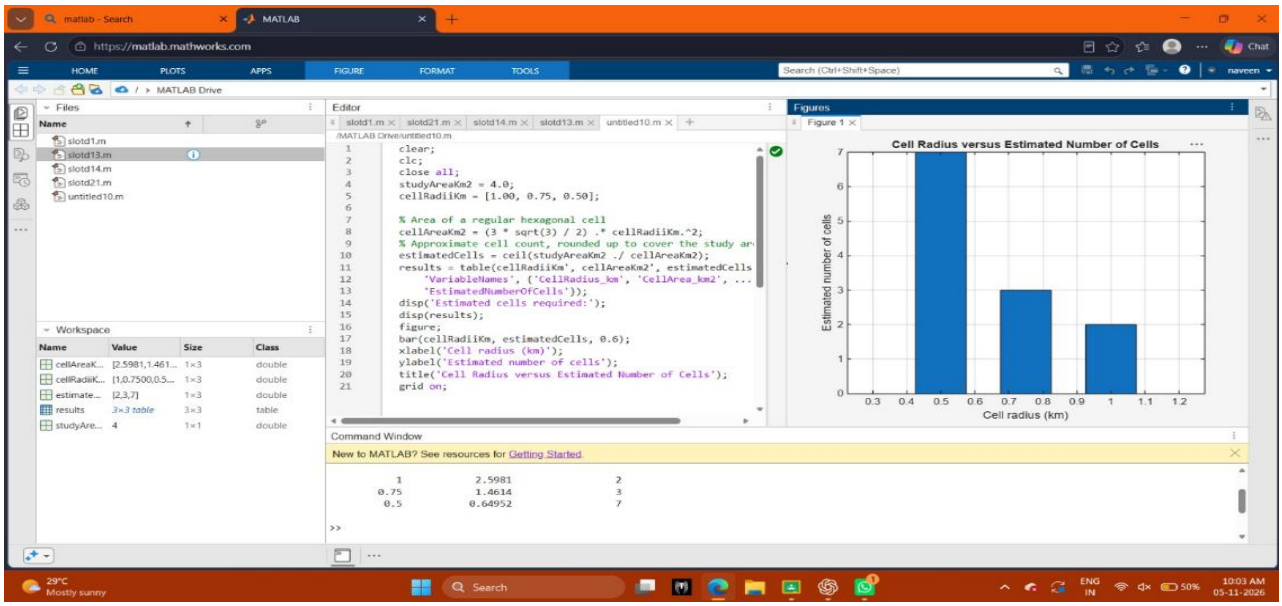
OBJECTIVE 2:

PROGRAM CODE :

```
clear;
clc;
close all;
studyAreaKm2 = 4.0;
cellRadiiKm = [1.00, 0.75, 0.50];

% Area of a regular hexagonal cell
cellAreaKm2 = (3 * sqrt(3) / 2) .* cellRadiiKm.^2;
% Approximate cell count, rounded up to cover the study area
estimatedCells = ceil(studyAreaKm2 ./ cellAreaKm2);
results = table(cellRadiiKm', cellAreaKm2', estimatedCells', ...
'VariableNames', {'CellRadius_km', 'CellArea_km2', ...
'EstimatedNumberOfCells'});
disp('Estimated cells required:');
disp(results);
figure;
bar(cellRadiiKm, estimatedCells, 0.6);
xlabel('Cell radius (km)');
ylabel('Estimated number of cells');
title('Cell Radius versus Estimated Number of Cells');
grid on;
```

EXPECTED OUTPUT :



OBJECTIVE 3 :

PROGRAM CODE :

```
clear;

clc;

close all;

studyAreaKm2 = 4.0;

userDensityPerKm2 = 30;

maxUsersPerCell = 50;

cellRadiiKm = [1.00, 0.75, 0.50];

totalDemand = studyAreaKm2 * userDensityPerKm2;

% Hexagonal cell area and approximate cell count

cellAreaKm2 = (3 * sqrt(3) / 2) .* cellRadiiKm.^2;

estimatedCells = ceil(studyAreaKm2 ./ cellAreaKm2);

% Estimate network capacity and users served

networkCapacity = estimatedCells .* maxUsersPerCell;

usersServed = min(totalDemand, networkCapacity);

results = table(cellRadiiKm', estimatedCells', networkCapacity', ...
```

usersServed', ...

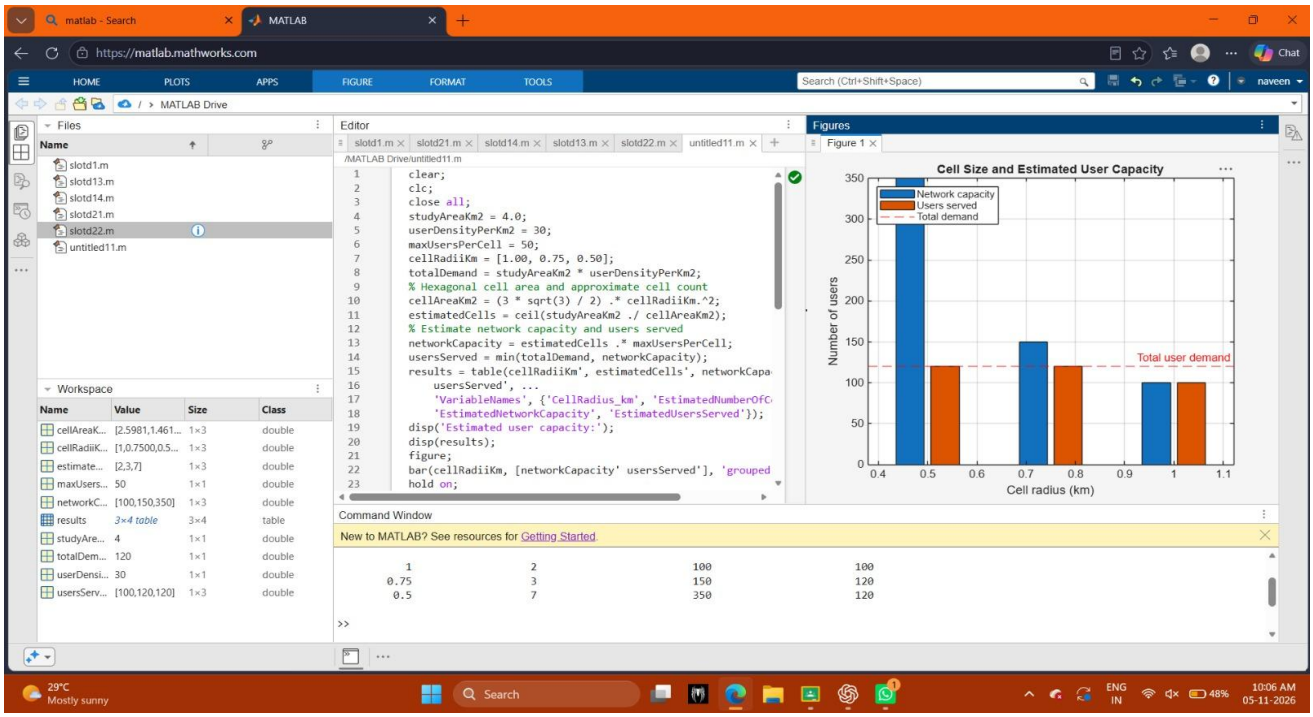
'VariableNames', {'CellRadius_km', 'EstimatedNumberOfCells', ...

'EstimatedNetworkCapacity', 'EstimatedUsersServed'});

disp('Estimated user capacity:');

disp(results);

EXPECTED OUTPUT:



OBJECTIVE 4 :

PROGRAM CODE :

clear;

clc;

close all;

cellRadiusKm = 0.5;

radiusM = cellRadiusKm * 1000;

% Centre-to-centre spacing of horizontally adjacent hexagonal cells

separationM = sqrt(3) * radiusM;

% Mobile user moves from base station 1 toward base station 2

xMeters = linspace(0, separationM, 201);

```
% Distance from the mobile to each base station
minimumDistanceM = 20; % Prevent log10(0) at a base-station location
distance1M = max(abs(xMeters), minimumDistanceM);
distance2M = max(abs(xMeters - separationM), minimumDistanceM);
% Illustrative log-distance path-loss parameters
transmitPowerdBm = 30;
pathLossAt1m_dB = 40;
pathLossExponent = 3.5;
% Estimate received power from each base station
receivedPower1dBm = transmitPowerdBm - pathLossAt1m_dB ...
- 10 * pathLossExponent .* log10(distance1M);
receivedPower2dBm = transmitPowerdBm - pathLossAt1m_dB ...
- 10 * pathLossExponent .* log10(distance2M);
% Handoff occurs when base station 2 becomes stronger
handoffIndex = find(receivedPower2dBm >= receivedPower1dBm, 1, 'first');
handoffPositionM = xMeters(handoffIndex);
fprintf('Estimated handoff position: %.1f m from base station 1\n', ...
handoffPositionM);
figure;

plot(xMeters / 1000, receivedPower1dBm, ...
'b', 'LineWidth', 1.5);
hold on;
plot(xMeters / 1000, receivedPower2dBm, ...
'r', 'LineWidth', 1.5);
xline(handoffPositionM / 1000, '--k', 'Estimated handoff');
xlabel('Mobile-user position (km)');
ylabel('Estimated received power (dBm)');
```

```
title('Received Signal Strength Between Adjacent Cells');
legend('Base station 1', 'Base station 2', ...
'Location', 'best');
grid on;
hold off;
```

EXPECTED OUTPUT :

